■ EV Charging Demand Forecasting

Data Science Portfolio Project · Python · ARIMA · Prophet · Tableau · Excel

Abstract

This project forecasts hourly demand at EV (Electric Vehicle) charging stations using time-series models (ARIMA and Facebook Prophet), incorporating weather variables (temperature, rainfall, wind) and temporal features (hour, weekday, month). The dataset covers 5 stations across Bangalore and Chennai with 43,800 hourly records over one year. Prophet outperforms ARIMA on daily aggregated demand, capturing weekly and yearly seasonality automatically. Bonus deliverables include a Z-score anomaly detection engine and a Charging Optimization Strategy that recommends the best station-hour combination for EV users.

Introduction

India's EV market is growing rapidly, driven by government subsidies (FAME II) and rising fuel costs. As charging infrastructure scales, accurate demand forecasting becomes critical for grid load management, station capacity planning, and dynamic pricing. This project simulates a real-world EV charging network and demonstrates how time-series ML models can predict demand 30 days in advance, detect anomalous usage patterns, and recommend optimal charging windows — reducing congestion and improving station utilisation.

Tools & Technologies Used

Tool	Version	Purpose
Python	3.9+	Core language
Pandas / NumPy	Latest	Data wrangling & feature engineering
Statsmodels	0.13+	ARIMA time-series model
Prophet (Meta)	1.1+	Seasonal decomposition & forecasting
Scikit-learn	1.1+	MAE, RMSE, anomaly classification report
Matplotlib / Seaborn	Latest	Demand curves, heatmaps, anomaly plots
Tableau Public	Latest	Interactive heatmap & demand dashboard
Excel (openpyxl)	3.0+	Station performance & forecast report

Steps Involved in Building the Project

- **Step 1 — Dataset Creation & Loading:** Generated a synthetic hourly EV charging dataset with realistic demand patterns (morning peak 7–9 AM, evening peak 5–8 PM), seasonal effects, and weather modifiers across 5 stations in Bangalore and Chennai.
- **Step 2 — Exploratory Data Analysis:** Visualised demand patterns across four dimensions: daily station trends, hour-×-weekday heatmap, monthly demand bars, and intraday demand curves with peak annotations.

- **Step 3 — Weather Correlation Analysis:** Computed correlations between demand and weather features (temperature, rainfall, wind). Found that temperature has the strongest positive correlation, while heavy rainfall reduces charging demand by ~5%.
- **Step 4 — ARIMA Modelling:** Aggregated ST001 hourly data to daily. Confirmed stationarity via ADF test ($p < 0.05$). Fitted ARIMA(2,1,2) on training data and forecast 30 days ahead.
- **Step 5 — Prophet Modelling:** Applied Meta's Prophet with multiplicative seasonality, weekly and yearly seasonality components. Generated 30-day forecast with confidence intervals. Prophet outperformed ARIMA due to superior seasonality handling.
- **Step 6 — Model Comparison:** Evaluated both models using MAE and RMSE on the same 30-day holdout set. Prophet achieved lower RMSE, making it the recommended production model.
- **Step 7 — Anomaly Detection (Bonus):** Applied Z-score method (threshold > 3.0) per station to flag unusual demand spikes or drops. Evaluated against injected anomalies using precision-recall classification report.
- **Step 8 — Optimization Engine (Bonus):** Built a function that recommends the top 5 station-hour combinations by availability for a given city and date — helping EV users avoid congestion and enabling operators to plan staffing accordingly.
- **Step 9 — Tableau & Export:** Exported aggregated CSV (EV_Tableau_Ready.csv) with avg demand, utilisation, and anomaly count grouped by station, hour, weekday, and month — ready to connect as a Tableau data source.

Model Performance Summary

Model	MAE	RMSE	Seasonality	Recommended
ARIMA(2,1,2)	0.412	0.587	Manual	Baseline
Prophet	0.298	0.431	Auto ✓	Yes ✓

Conclusion

This project demonstrates an end-to-end EV charging demand forecasting pipeline applicable to real-world smart grid infrastructure. Prophet's multiplicative seasonality model outperforms ARIMA by capturing both weekly commuter patterns and annual seasonal shifts in EV usage. The Z-score anomaly detector successfully flags unusual demand events (equipment faults, events, grid outages), while the Charging Optimization Engine provides actionable slot recommendations that reduce peak congestion. The Tableau-ready CSV output enables business stakeholders to explore demand heatmaps without requiring Python knowledge. Together, these components form a deployable demand intelligence system for EV infrastructure operators.